2.1 Maxwell's Equations

For the electromagnetic field, the interaction is generally divided into three contributions: the action of the charged particle itself S_m, the interaction action between the charged particle and the field S_{fm}, and the action of the field itself S_f. When studying the trajectory of a charged particle, we need to vary the particle's action S_m and the particle-field interaction action S_{fm}. When studying the field equations governing the field itself, we need to vary the interaction action S_{fm} and the field action S_f.

We now derive the equation of motion for a point charge in an electromagnetic field. First, we construct the action for this system, namely,

$$S = S_m + S_{fm} \tag{2.1.1}$$

Substituting Eq. (1.3.18) into Eq. (2.1.1), we obtain

$$S = \int -mc^2 \sqrt{1 - \left(\frac{v}{c}\right)^2}\, dt + S_{fm} \tag{2.1.2}$$

Examining Eq. (2.1.2), we see that the interaction term S_{fm} must also take a scalar form, just like the other terms in the action S_m. Since this term describes the interaction between the charge and the electromagnetic field, we express the electromagnetic potentials in four-dimensional form, denoted by $A^i = (\varphi, \boldsymbol{A})$. The simplest scalar expression we can construct is therefore

$$S_{fm} = -\frac{e}{c}\int A^i dx_i \tag{2.1.3}$$

Combining Eqs. (2.1.2) and (2.1.3), we obtain

$$\begin{aligned} S &= \int -mc^2 \sqrt{1 - \left(\frac{v}{c}\right)^2}\, dt - \frac{e}{c}\int A^i dx_i \\ &= \int -mc^2 \sqrt{1 - \left(\frac{v}{c}\right)^2}\, dt - \int e\varphi dt + \frac{e}{c}\int A^i \frac{dx^i}{dt} dt \end{aligned} \tag{2.1.4}$$

Rearranging Eq. (2.1.4) and using Eq. (1.2.1), we derive the Lagrangian for a charged particle in an electromagnetic field:

$$L = -mc^2\sqrt{1-\left(\frac{v}{c}\right)^2} - e\varphi + \frac{e}{c}\boldsymbol{A}\cdot\boldsymbol{v} \tag{2.1.5}$$

Substituting Eq. (2.1.5) into the equation of motion for a charged particle, Eq. (1.2.3), we obtain

$$\frac{\partial L}{\partial \boldsymbol{r}} - \frac{d}{dt}\left(\frac{\partial L}{\partial \boldsymbol{v}}\right) = -e\cdot grad\varphi + \frac{e}{c}\cdot grad(\boldsymbol{A}\cdot\boldsymbol{v}) - \frac{d}{dt}\left[\boldsymbol{p} + \frac{e}{c}\boldsymbol{A}\right] \tag{2.1.6}$$

To evaluate the second term on the right-hand side of Eq. (2.1.6), we need to expand it using vector calculus, namely,

$$\begin{aligned} grad(\boldsymbol{A}\cdot\boldsymbol{B}) &= \frac{\partial}{\partial x_i}\left(a_j b_j\right) = \frac{\partial a_j}{\partial x_i}b_j + a_j\frac{\partial b_j}{\partial x_i} \\ &= \delta_{ik}\delta_{jl}\frac{\partial a_l}{\partial x_k}b_j + \delta_{ik}\delta_{jl}a_j\frac{\partial b_l}{\partial x_k} \\ &= \left(\varepsilon_{mij}\varepsilon_{mkl} + \delta_{il}\delta_{jk}\right)\left(\frac{\partial a_l}{\partial x_k}b_j + a_j\frac{\partial b_l}{\partial x_k}\right) \\ &= -\varepsilon_{imj}\varepsilon_{mkl}\frac{\partial a_l}{\partial x_k}b_j + \frac{\partial a_i}{\partial x_k}b_k + \varepsilon_{ijm}a_j\varepsilon_{mkl}\frac{\partial b_l}{\partial x_k} \\ &\quad + a_k\frac{\partial b_i}{\partial x_k} \\ &= \boldsymbol{B}\times\boldsymbol{rot}\,\boldsymbol{A} + (\boldsymbol{B}\cdot\boldsymbol{\nabla})\cdot\boldsymbol{A} + \boldsymbol{A}\times\boldsymbol{rot}\,\boldsymbol{B} + (\boldsymbol{A}\cdot\boldsymbol{\nabla})\cdot\boldsymbol{B} \end{aligned} \tag{2.1.7}$$

Substituting Eq. (2.1.7) into Eq. (2.1.6), and noting that the velocity $\boldsymbol{v}$ is independent of the spatial coordinates $\boldsymbol{r}$, we obtain

$$-e\cdot grad\varphi + \frac{e}{c}\cdot[\boldsymbol{v}\times\boldsymbol{rot}\,\boldsymbol{A} + (\boldsymbol{v}\cdot\boldsymbol{\nabla})\cdot\boldsymbol{A}] - \frac{d\boldsymbol{p}}{dt} - \frac{e}{c}\frac{d\boldsymbol{A}}{dt} = 0 \tag{2.1.8}$$

The vector potential $\boldsymbol{A}$ has the general form $\boldsymbol{A} = \boldsymbol{A}(\boldsymbol{r}, t)$, and therefore its total time t derivative can be decomposed as

$$\frac{d\boldsymbol{A}}{dt} = \frac{\partial \boldsymbol{A}}{\partial t} + \frac{\partial \boldsymbol{A}}{\partial \boldsymbol{r}}\frac{d\boldsymbol{r}}{dt} = \frac{\partial \boldsymbol{A}}{\partial t} + (\boldsymbol{v}\cdot\boldsymbol{\nabla})\cdot\boldsymbol{A} \tag{2.1.9}$$

Substituting Eq. (2.1.9) into Eq. (2.1.8), we obtain

$$\frac{d\boldsymbol{p}}{dt} = -e\cdot grad\varphi - \frac{e}{c}\frac{\partial \boldsymbol{A}}{\partial t} + \frac{e}{c}\cdot(\boldsymbol{v}\times\boldsymbol{rot}\,\boldsymbol{A}) \tag{2.1.10}$$

We define

$$\boldsymbol{E} = -grad\varphi - \frac{1}{c}\frac{\partial \boldsymbol{A}}{\partial t},\quad \boldsymbol{H} = \boldsymbol{rot}\,\boldsymbol{A} \tag{2.1.11}$$

Substituting Eq. (2.1.11) into Eq. (2.1.10) gives

$$\frac{d\boldsymbol{p}}{dt} = e\cdot\boldsymbol{E} + \frac{e}{c}\cdot(\boldsymbol{v}\times\boldsymbol{H}) \tag{2.1.12}$$

Equation (2.1.12) is the equation of motion for a charged particle in an

electromagnetic field. By examining the electric field strength $\boldsymbol{E}$ and the magnetic field strength $\boldsymbol{H}$, we can readily obtain the first pair of Maxwell's equations:

$$\boldsymbol{rot}\,\boldsymbol{E} = -\frac{1}{c}\frac{\partial \boldsymbol{H}}{\partial t}, \qquad div\,\boldsymbol{H} = 0 \tag{2.1.13}$$

We now return to the expression for the action in Eq. (2.1.4) and rewrite it slightly to obtain

$$S = \int -mcds - \frac{e}{c}\int A^i dx_i \tag{2.1.14}$$

We then vary Eq. (2.1.14), obtaining

$$\begin{aligned}\delta S = \delta\left(\int -mcds - \frac{e}{c}\int A^i dx_i\right) &= \delta\left(\int -mc\sqrt{dx^i dx_i} - \frac{e}{c}\int A^i dx_i\right)\\ &= -mc\int \frac{dx^i\cdot\delta dx_i + dx_i\cdot\delta dx^i}{2\sqrt{dx^i dx_i}} - \frac{e}{c}\int \delta A^i dx_i\\ &\quad -\frac{e}{c}\int A^i\cdot\delta dx_i\end{aligned} \tag{2.1.15}$$

Using the commutativity of variation and differentiation, we obtain

$$\delta S = -mc\int \frac{dx^i}{ds} d\delta x_i - \frac{e}{c}\int \delta A^i dx_i - \frac{e}{c}\int A^i d\delta x_i \tag{2.1.16}$$

For Eq. (2.1.16), we introduce the four-velocity, defined as

$$u^i = \frac{dx^i}{ds} \tag{2.1.17}$$

Substituting this definition into Eq. (2.1.16), we have

$$\begin{aligned}\delta S = -\int mcu^i d\delta x_i - \frac{e}{c}\int \delta A^i dx_i - \frac{e}{c}\int A^i d\delta x_i &= -\left(mcu^i + \frac{e}{c}A^i\right)\delta x_i\Big|_a^b + \int \delta x_i d\left(mcu^i + \frac{e}{c}A^i\right)\\ &\quad -\frac{e}{c}\int \delta A^i dx_i\\ &= 0 + \int \delta x_i d\left(mcu^i + \frac{e}{c}A^i\right) - \frac{e}{c}\int \delta A^k dx_k\end{aligned} \tag{2.1.18}$$

Here, the differential dA^i and the variation δA^k of the four-potential can be expressed as

$$dA^i = \frac{\partial A^i}{\partial x_k} dx_k, \qquad \delta A^k = \frac{\partial A^k}{\partial x_i}\cdot\delta x_i \tag{2.1.19}$$

Substituting Eq. (2.1.19) into Eq. (2.1.18), we obtain

$$\delta S = \int \delta x_i \left[mc \frac{\partial u^i}{\partial x_k} + \frac{e}{c} \left(\frac{\partial A^i}{\partial x_k} - \frac{\partial A^k}{\partial x_i} \right) \right] dx_k = 0 \tag{2.1.20}$$

Since the variations δx_i are arbitrary, it follows that

$$mc du^i = \frac{e}{c} \left(\frac{\partial A^k}{\partial x_i} - \frac{\partial A^i}{\partial x_k} \right) dx_k \tag{2.1.21}$$

Dividing both sides of Eq. (2.1.21) by the spacetime line element ds, we obtain

$$mc \frac{du^i}{ds} = \frac{e}{c} \left(\frac{\partial A^k}{\partial x_i} - \frac{\partial A^i}{\partial x_k} \right) u_k \tag{2.1.22}$$

We now introduce a new second-rank four-dimensional tensor, defined as follows:

$$F^{ik} = \frac{\partial A^k}{\partial x_i} - \frac{\partial A^i}{\partial x_k} \tag{2.1.23}$$

We refer to the tensor F^{ik} as the electromagnetic field tensor. Each of its components corresponds to a component of the electric field strength $\boldsymbol{E}$ or the magnetic field strength $\boldsymbol{H}$. Moreover, it is an antisymmetric tensor with 6 independent components. Equation (2.1.22) can therefore be rewritten as

$$mc \frac{du^i}{ds} = \frac{e}{c} F^{ik} u_k \tag{2.1.24}$$

This is the equation of motion for a charged particle in an electromagnetic field, expressed in four-dimensional form.

To obtain the second pair of Maxwell's equations, we can make use of the Euler–Lagrange equation for local field theory, Eq. (1.4.8). However, the Lagrangian density is not yet known, so we must first determine its specific form before substituting it into Eq. (1.4.8). Depending on the choice of gauge, the four-potential of the field, denoted by A^i, can take different forms. For example, when solving problems involving weak gravitational fields at low velocities, we often choose an equipotential surface as the reference level at which the gravitational potential is set to zero. The issue of gauge choice will be discussed later in this work and need not be considered here. For now, it is sufficient to note that changing the gauge leads to a corresponding change in the potential.

However, the field equations we seek should be valid for all gauge choices. We

therefore need to work with physical quantities that remain invariant under changes in the gauge (or, equivalently, in the choice of potentials). Examples of such quantities are the electric field strength $\boldsymbol{E}$ and the magnetic field strength $\boldsymbol{H}$. Based on Eq. (2.1.23), the discussion that follows, and the requirement of covariance of the tensor equations, we can infer that the Lagrangian density of the electromagnetic field should be a scalar constructed from the electromagnetic field tensor F^{ik}. Since field Lagrangian densities are generally quadratic in the field variables, we propose that the electromagnetic field Lagrangian density takes the form

$$\mathcal{L}_f = F^{ik}F_{ik} = \left(\frac{\partial A^k}{\partial x_i} - \frac{\partial A^i}{\partial x_k}\right) \cdot \left(\frac{\partial A_k}{\partial x^i} - \frac{\partial A_i}{\partial x^k}\right) \tag{2.1.25}$$

In this case, the quantity φ_a in Eq. (1.4.8) corresponds to the four-potential A^i. We must also consider the interaction between the charge and the field. The total action should therefore be written as

$$S = -\frac{e}{c}\int A_i dx^i + \alpha \int \mathcal{L}_f d\Omega \tag{2.1.26}$$

where α is a constant coefficient.

We now need to express the interaction term in terms of a density. Considering a charge density ρ corresponding to a charge e, we have

$$\begin{aligned} S_{fm} &= -\int \frac{\rho}{c} dV \int A_i dx^i = -\int \frac{\rho}{c^2} dV \int A_i \frac{dx^i}{dt} c dt \\ &= -\int \frac{\rho}{c} A_i \frac{dx^i}{cdt} (cdtdV) \end{aligned} \tag{2.1.27}$$

Combining this result with Eq. (1.4.5), we obtain

$$S_{fm} = -\int A_i \frac{1}{c^2}\left(\rho \frac{dx^i}{dt}\right) d\Omega \tag{2.1.28}$$

We now introduce a new four-vector, namely the expression enclosed in parentheses in Eq. (2.1.28), which is defined as

$$j^i = \rho \frac{dx^i}{dt} \tag{2.1.29}$$

We refer to this quantity as the four-dimensional current density. Substituting it into Eq. (2.1.26), we obtain

$$S = \int \left(\alpha \mathcal{L}_f - \frac{A_i j^i}{c^2} \right) d\Omega \tag{2.1.30}$$

The total Lagrangian density is therefore given by the expression in parentheses in Eq. (2.1.30). Substituting it into the Euler–Lagrange equation (1.4.8), we obtain

$$\begin{aligned} \frac{\partial \left(\alpha \mathcal{L}_f - \frac{A_i j^i}{c^2} \right)}{\partial A_k} = -\frac{j^k}{c^2} &= \frac{\partial}{\partial x^i} \left[\frac{\partial \left(\alpha \mathcal{L}_f - \frac{A_i j^i}{c^2} \right)}{\partial \left(\frac{\partial A_k}{\partial x^i} \right)} \right] \\ &= 2\alpha \left(\frac{\partial^2 A^k}{\partial x^i \partial x_i} - \frac{\partial^2 A^i}{\partial x^i \partial x_k} \right) \end{aligned} \tag{2.1.31}$$

For $k = 0$, we have

$$\begin{aligned} -\frac{j^0}{c^2} &= 2\alpha \left(\frac{\partial^2 \varphi}{\partial x^i \partial x_i} - \frac{\partial^2 A^i}{\partial x^i \partial x_0} \right) \\ &= 2\alpha \left[\frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} - div\ (grad\ \varphi) - \frac{1}{c} \frac{\partial}{\partial t} \left(\frac{1}{c} \frac{\partial \varphi}{\partial t} + div\ \boldsymbol{A} \right) \right] \\ &= 2\alpha \cdot div\ \left(-grad\ \varphi - \frac{1}{c} \frac{\partial \boldsymbol{A}}{\partial t} \right) \end{aligned} \tag{2.1.32}$$

Substituting Eq. (2.1.11) into Eq. (2.1.32), we obtain

$$div\ \boldsymbol{E} = -\frac{j^0}{2\alpha c^2} = -\frac{\rho}{2\alpha c} \tag{2.1.33}$$

For $k = 1,2,3$, Eq. (2.1.31) can be rewritten in vector form as

$$-\frac{\boldsymbol{j}}{c^2} = 2\alpha \left[\frac{1}{c^2} \frac{\partial^2 \boldsymbol{A}}{\partial t^2} + \frac{1}{c} \frac{\partial}{\partial t} (grad\ \varphi) + \boldsymbol{rot}\ (\boldsymbol{rot}\ \boldsymbol{A}) \right] \tag{2.1.34}$$

Since $\boldsymbol{j} = \rho \boldsymbol{v}$, this gives

$$\boldsymbol{rot}\ \boldsymbol{H} = \frac{1}{c} \frac{\partial \boldsymbol{E}}{\partial t} - \frac{\boldsymbol{j}}{2\alpha c^2} \tag{2.1.35}$$

Taking $\alpha = -1/8\pi c$ and substituting this value into Eqs. (2.1.33) and (2.1.35), we obtain

$$\begin{cases} div\ \boldsymbol{E} = \dfrac{4\pi j^0}{c} = 4\pi\rho \\ \boldsymbol{rot}\ \boldsymbol{H} = \dfrac{1}{c} \dfrac{\partial \boldsymbol{E}}{\partial t} + \dfrac{4\pi \boldsymbol{j}}{c} \end{cases} \tag{2.1.36}$$

Together with Eq. (2.1.13), Eq. (2.1.36) completes the set of Maxwell's equations:

$$\begin{cases} div\,\boldsymbol{E} = 4\pi\rho \\ \boldsymbol{rot}\,\boldsymbol{E} = -\dfrac{1}{c}\dfrac{\partial \boldsymbol{H}}{\partial t} \\ div\,\boldsymbol{H} = 0 \\ \boldsymbol{rot}\,\boldsymbol{H} = \dfrac{1}{c}\dfrac{\partial \boldsymbol{E}}{\partial t} + \dfrac{4\pi \boldsymbol{j}}{c} \end{cases} \tag{2.1.37}$$

Maxwell's equations occupy an extraordinarily important position in modern physics. They represent a decisive triumph of the field concept, unifying the electric and magnetic fields within a single theoretical framework. They also inspired Einstein's subsequent thinking about the gravitational field.